

from outside, but we manufacture the gate-phase part for it. We have our own embedded team. Design-to-finish hardware is done in-house. PCB design and software are also done in-house. Only PCB fabrication is outsourced,” says Thangathurai.

For battery management, they do not have backup inside the device yet. In industrial settings, the inverter supply is given externally to feed the equipment. In the near future, they plan to add built-in battery backup. For data, they have already achieved real-time updates. The device sends data every 500 milliseconds.

The main challenge they face now is not technology but market entry. As a new company, the difficulty lies in customer acquisition. It is mainly a marketing challenge and they are working through it. On discussing the revenue generated, Thangathurai discloses, “The product line was released only one year ago. To date, approximately 100 units have been sold. We know this is not enough and we are now working to scale and reach good volume.”

A new line has also started under the name Veoma Track.

Veoma means wind. “Under this line we are introducing RFID technology for tracking. One RFID model is for warehouse management, and another is for jewellery shops. These are our own products and both are currently in the pilot stage,” adds Thangathurai.

For future growth, the startup plans to expand primarily through marketing partnerships, not heavy internal hiring. They are looking for dealers, distributors, and committed partners who can take the products to the market.

—Nidhi Agarwal

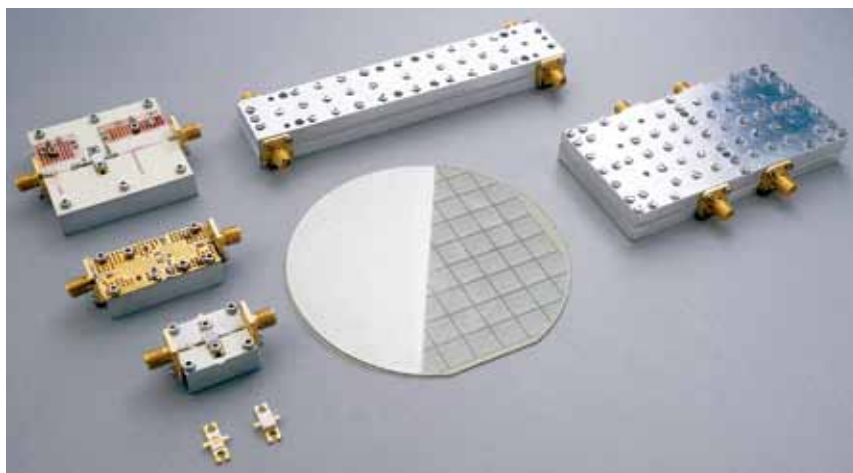
2 FROM LAB TO FAB: HOW AGNIT IS MAKING GALLIUM NITRIDE CHIPS IN INDIA

The second wave of Covid-19 was an unlikely moment to launch a gallium nitride chip, yet AGNIT converted the challenge into momentum and built its first working prototypes within two years. Co-founder Hareesh Chandrasekar notes that the early progress stemmed from disciplined execution during the period of uncertainty. From its pilot facility at IISc, the company runs a vertically integrated line that spans wafer growth to final testing.

Founded in 2019 and incubated at the Indian Institute of Science (IISc), Bengaluru, AGNIT Semiconductor positions itself as a vertically integrated GaN semiconductor company. The company designs and makes its own wafers, processing them, dicing the chip, assembling, packing, and testing them. Each chip is designed to meet different specifications and is supplied to customers.

“Having technology is one thing, but commercialising it to solve specific customer pain points is quite another,” Chandrasekar says. AGNIT’s markets include defence, space, and telecom, with consumer-facing applications currently under development.

Intellectual property is a central component of the model. The company licenses background IP from IISc and builds its own portfolio, with six applications filed in 2023. Chandrasekar says,



Products offered by AGNIT Semiconductor

“Intellectual property for us is not just patents, it is also trade secrets.” By combining licensed research from IISc with its filings, the company reflects a broader effort to shift India’s role in semiconductors from design services to ownership of core technology.

The company has been operating

since the Covid-19 pandemic. Chandrasekar adds, “It was a hard time, especially the second wave. Thankfully, the team stayed together. The period was spent developing, filing patents, and building prototypes.”

By 2022, AGNIT had shipped its first GaN wafers and developed initial RF amplifier prototypes,